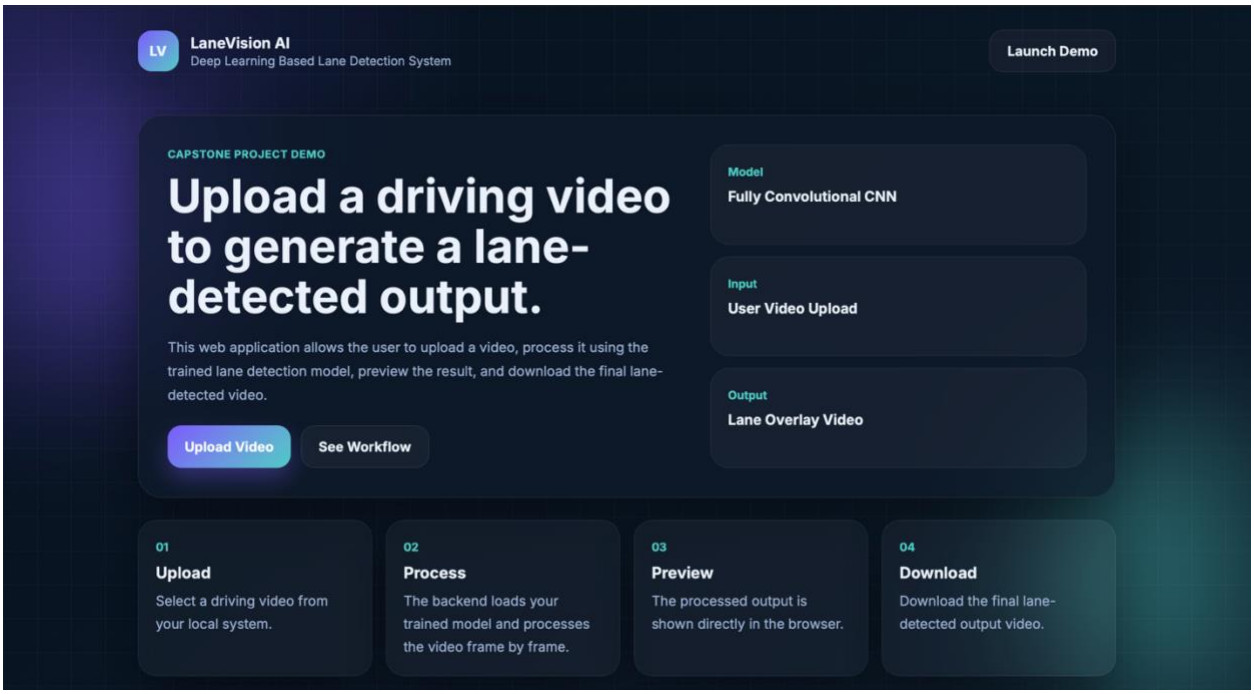
CAPSTONE PROGRESS REPORT 2

Project Title: Lane Detection Using Deep Learning
Course: DATS 6501 – Data Science Capstone

Summary

This project is about building a lane detection system for driving videos using a deep learning approach. Since the first progress report, the focus of the project has shifted from the actual development of the models and video testing to a more complete end-to-end application.

The key accomplishments of the second phase of the project were the successful implementation of a web-based application that allows a user to upload a video file, process the video with the trained model, display the processed video, and download the final output video file.



Work Completed Since Progress Report 1

A. Model Pipeline

- Preserved the trained model in .h5 format for reuse in inference.
- Processed uploaded videos frame by frame using the trained model.
- Generated annotated output videos with detected lane overlays.

B. Web Application Development

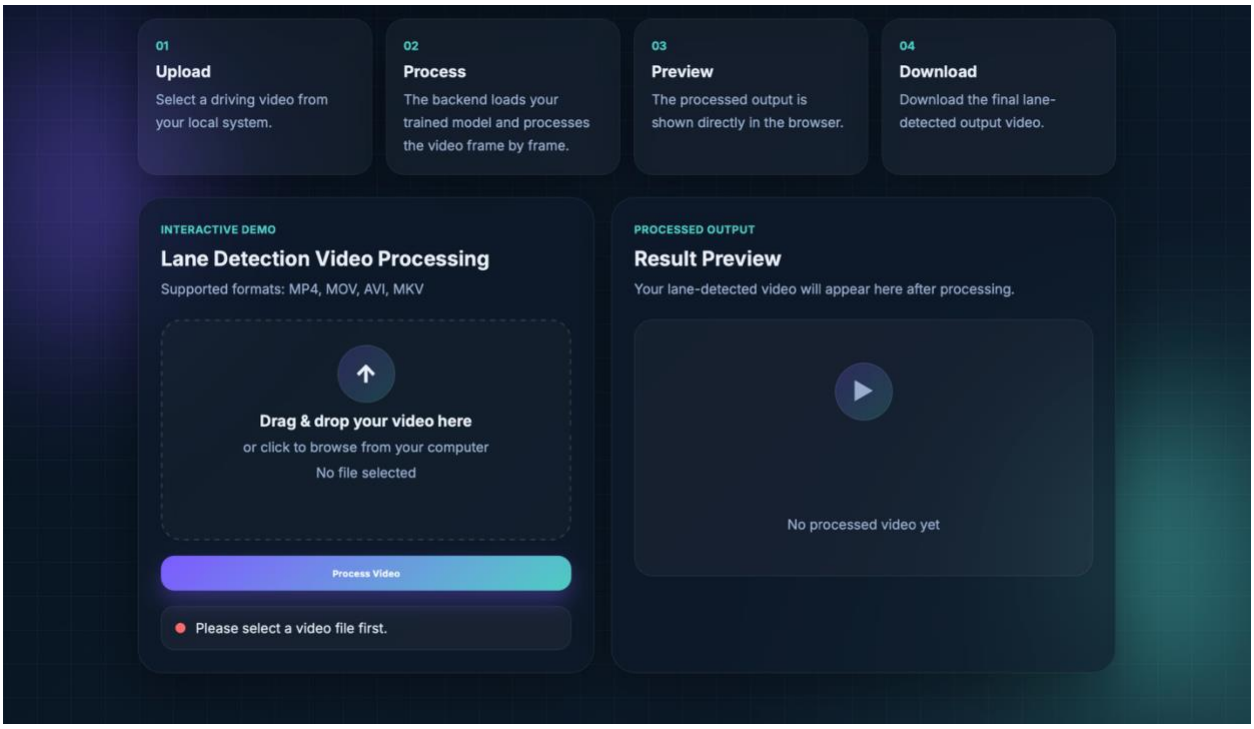
- Built a Flask based backend to connect the model with the user interface.
- Added support for video upload and file type validation.
- Integrated backend processing to run lane detection on uploaded videos.
- Enabled output preview and download functionality for processed videos.

C. Frontend Interface and User Workflow

- Added a landing page that explains the project clearly.
- Created an upload section where users can drag and drop or browse for videos.
- Structured the workflow as: Upload → Process → Preview → Download.

D. Improved Presentation Quality

- Designed the application with a cleaner and more professional layout.
- Added status messages to improve user understanding during processing.
- Improved the overall demo quality compared to the earlier command-line version.



Current System Workflow

At the current stage, the system processes in the following steps:

1. The user accesses the Lane Vision AI web application.
2. The user uploads a driving video from the local machine.

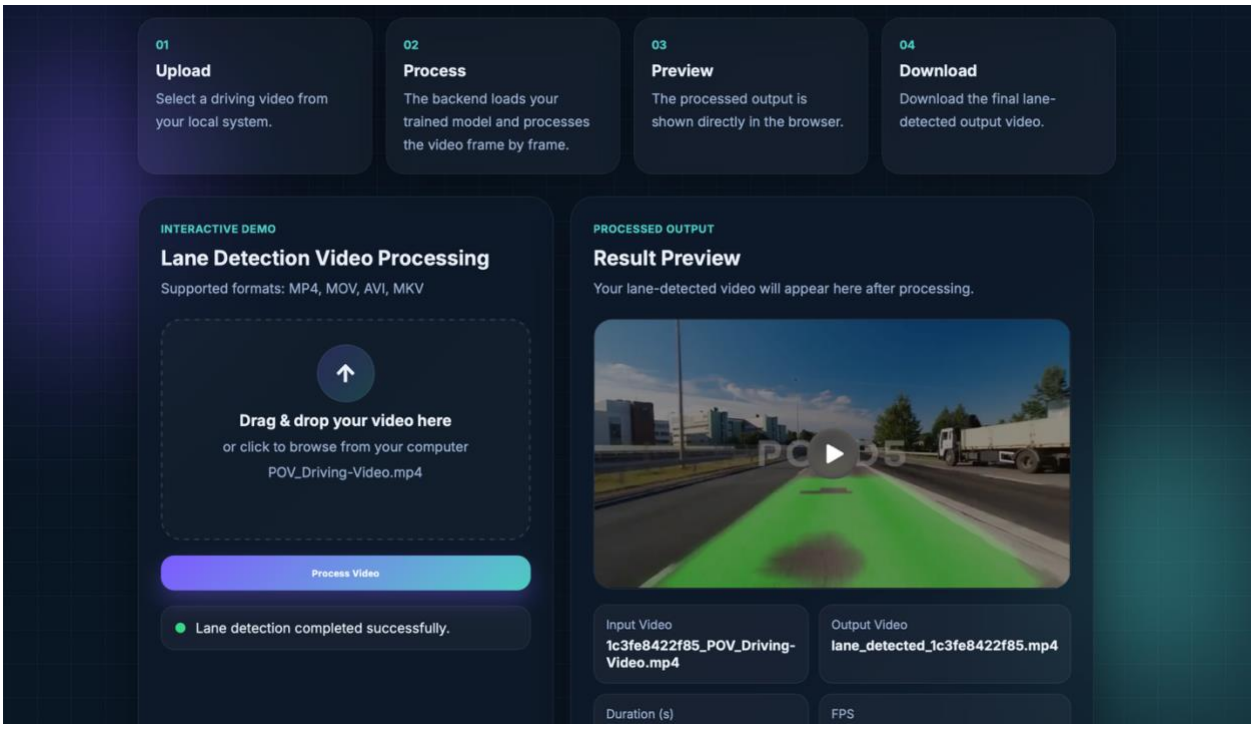
3. The backend checks the file type and saves the uploaded video.
4. The uploaded video is processed frame by frame by the pre-trained FCN model.
5. The predicted lanes are overlaid on the original video frames to obtain the final processed video.
6. The processed video is displayed in the browser.
7. The user can download the final processed video with lane detection.

Current Results and Demonstration Status

• The system is now able to handle a user-uploaded driving video via the web application and produce a lane detected output video.

• The current demo presents an entire workflow with three stages: a landing page, an upload/processing page, and a preview page.

• The video is now playable directly in the browser and downloadable, which makes this project more useful compared to the previous version.



Challenges Encountered

• The accuracy of lane detection may still be questionable under unfavorable driving conditions such as shadows, curves, faded lanes, and varying lighting.

- The implementation of the deep learning model in a web application was another challenge, considering the handling of file upload and output.
- The maintenance of a smooth user interface during video processing is another challenge.
- The above challenges pointed out the areas that need to be improved before the final submission.

Next Development Milestones

- Improve the performance of the model in difficult road and lighting conditions.
- Add better evaluation metrics like IoU and F1-score to evaluate the performance of the system.
- Enhance the web interface to show better processing information and output visualization.

Conclusion

In summary, the project has moved from model development to a working web-based lane detection system. This means a user can upload a driving video, process it, preview the output, and finally download the output. The next steps will be on improving performance, evaluating, and making it ready for final presentation.